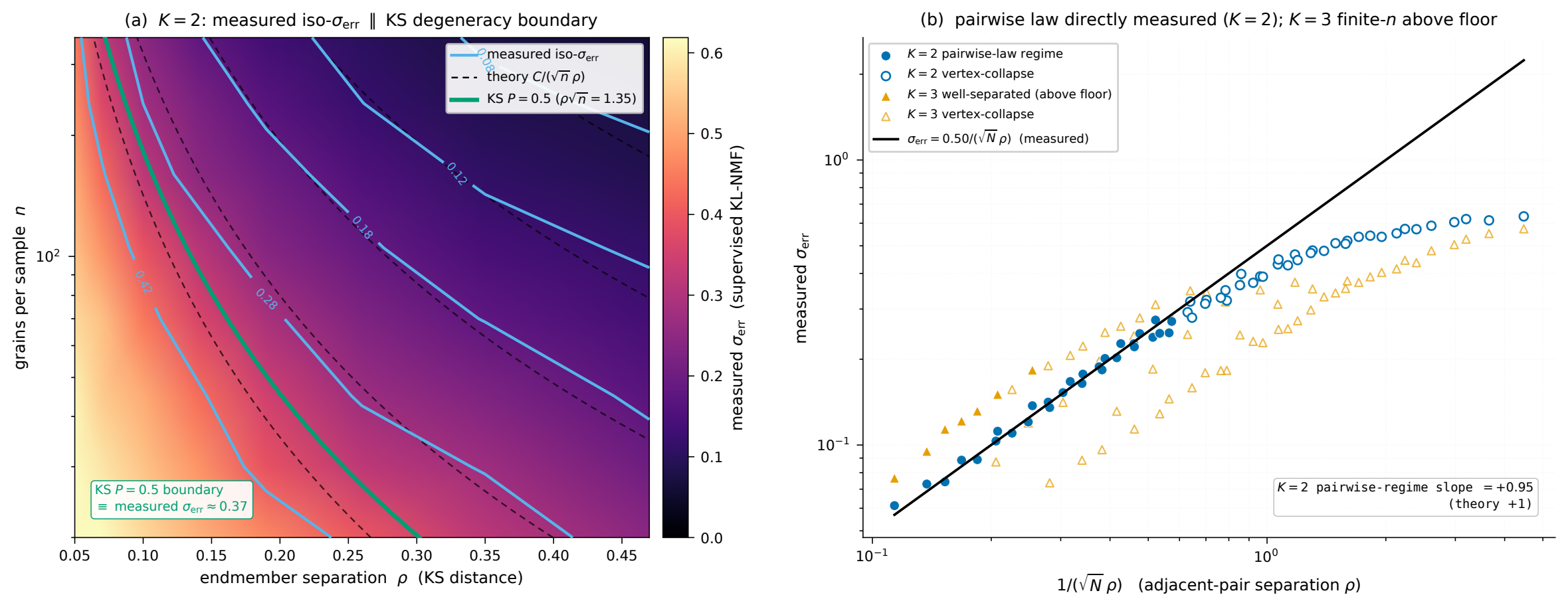
(a) $K = 2$: measured iso-σ_{err} ∥ KS degeneracy boundary
measured iso-σ_{err}
theory $C/(\sqrt{n}\,\rho)$
KS $P = 0.5$ ($\rho\sqrt{n} = 1.35$)
0.08
0.12
0.18
0.28
0.42
KS $P = 0.5$ boundary
≡ measured $\sigma_{\mathrm{err}} \approx 0.37$
grains per sample n
10^2
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45
endmember separation ρ (KS distance)
measured σ_{err} (supervised KL-NMF)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
(b) pairwise law directly measured ($K = 2$); $K = 3$ finite-n above floor
$K = 2$ pairwise-law regime
$K = 2$ vertex-collapse
$K = 3$ well-separated (above floor)
$K = 3$ vertex-collapse
$\sigma_{\mathrm{err}} = 0.50/(\sqrt{N}\,\rho)$ (measured)
$K = 2$ pairwise-regime slope = +0.95
(theory +1)
measured σ_{err}
10^0
10^{-1}
10^{-1}
10^0
$1/(\sqrt{N}\,\rho)$ (adjacent-pair separation ρ)